Technology, Automation, and Indian Labour Markets

Research Memo | Abhishek Gupta

1. Research Question

Which Indian states and occupations are most exposed to technology and automation, and does exposure relate to employment and wage outcomes, particularly for women and lower-skilled workers?

2. Data and Method

We combine three rounds of unit-level PLFS data (2022–2024) with text-as-data exposure indices. Using the task-based framework (Autor, Levy, Murnane 2003), we parse NCO-2015 occupation descriptions, tag them using ALM routine/non-routine keywords, and calculate a `Routine Intensity` score (0 to 1) for each 3-digit occupation. We merge this to 131,834 salaried workers across 34 major states. We estimate log-wage models with State and Year fixed effects, clustered by state.

3. Preliminary Evidence

Finding 1: Routine exposure carries a heavy wage penalty, especially for women.

Our baseline OLS specification shows that a 1-unit increase in Routine Intensity reduces male wages by 59% ($p<0.01$). However, the interaction term (Routine × Female) is -0.347 ($p<0.01$). This means women in highly routine jobs face an *additional* 35% wage penalty compared to men in the same jobs.

Finding 2: The penalty is worse for high-skilled and rural women.

Contrary to standard ALM theory (which predicts low-skill workers suffer most), our heterogeneity splits show that the routine penalty for women is larger for high-skilled workers (-0.40) than for low-skilled workers (-0.32). This suggests educated women are trapped in routine clerical or data-entry roles that are heavily devalued, while low-skilled women are already at a wage floor.

Finding 3: Machine Learning confirms the econometric results.

We trained an XGBoost model to predict wages (Out-of-sample $R^2 = 0.4568$). Using SHAP (SHapley Additive exPlanations) to interpret the black-box model, we find that `Routine_Intensity` and `Female` are the top two most important predictors of wage suppression in the dataset. The ML model perfectly aligns with the linear regression, confirming the robustness of the econometric findings.

4. Limitations

- **Reverse Causality:** We observe cross-sectional correlation, not causality. Low-wage states may adopt routine technologies faster, or routine jobs may simply pay less.

- **Exposure Construction**: The text-based exposure index is a proxy based on keyword matching; it misses context. (e.g., "operating a machine" vs "designing a machine").

- **Small Clusters:** We have ~34 state clusters. We use clustered standard errors, but wild-bootstrap methods would be more robust.

- **2025 Exclusion:** PLFS revamped its sampling methodology in 2025, so we restricted the panel to 2022–2024 for comparability.

5. Next Steps

Future work will incorporate job posting data to measure actual firm-level adoption and a primary survey to validate the exposure scores directly with workers I will also incorporate a full literature review.